

Feature Status — against Cindi's App Feature Proposal

A line-by-line trace of the requirements document (2026-05-20) to what exists in the app today. Use this to walk through the proposal in a meeting and see, for each item, whether it is done, partly done, or still open.

Legend:  done ·  partly ·  not started

The skill chain

The training sequence, and where the app now sits:

#	Step	What it teaches	Status
1	Measure my crossing	How long your crossing actually takes	 New
2	Practise my timing	An intuitive, felt sense of that duration	 Existing
3	Compare practice	Judging a gap against your crossing time	 New
4	At the street	The same judgement, with real vehicles	 New

Step 1 was the gap: the app previously *began* from two numbers an instructor measured with a stopwatch and typed into Settings. It now measures them itself.

User interface

Item	Status	Notes
Divide up menus / settings for easier navigation	✓	Home screen with four numbered steps; each activity is its own screen. Settings and Accessibility stay as dialogs.

Sounds

Item	Status	Notes
Refine within/outside margin sounds — not easily identifiable, not loud enough at a roadside	✓	Rebuilt. They differ on three independent axes: pitch register, duration (0.12 s vs 0.30 s), and texture (pure chime vs two close frequencies beating). Outside cue moved from 240–260 Hz to 294–349 Hz because phone speakers roll off below ~300 Hz. Debug volumes that were clipping at 4× gain were reset and a master limiter added. Still needs a listen at a real roadside.
Allow users to customize sounds	●	Two presets per role, selectable with previews. Arbitrary user-supplied sounds are not supported.

Settings

Item	Status	Notes
Instructor sets margin and crossing times remotely via emailed/texted link	✓	Settings → Share time settings. Generates a checksummed link plus a separate plain link, so rich previews stay readable.
Keep that link mechanism working	✓	Carried over unchanged.

Button used for practice

Item	Status	Notes
Build feedback into the button itself — different colour/shape	✓	The action button encodes stage four ways at once: colour, corner radius, border style, and glyph. Never colour alone.
Haptics/vibration for DeafBlind users, or noisy situations	✓	Native iOS Taptic Engine via Capacitor; Web Vibration API on Android; degrades silently in iOS Safari, which has never supported it. Distinct patterns for marker, within-margin, and outside-margin.

Tutorial / Help

Item	Status	Notes
HELP section — instructors		Help screen, “For instructors”: setup, margin guidance, feedback modes, progress, roadside noise sampling.
HELP section — clients		Help screen, “For clients”: the four steps in plain language.
Tutorial on first open, re-runnable		Runs automatically on first launch; re-runnable from Help. Plays each sound with an explanation of what it means.

Practice Comparison Task

Item	Status	Notes
Separate from / after developing timing	✓	Step 3, gated until crossing times exist.
User chooses half street or full street	✓	
Continuous sound option	✓	Default.
Two sounds with silence between	✓	Selectable — you were unsure which reads better, so both ship.
Answer longer / shorter / about the same	✓	Three full-width buttons, minimum 4.2 rem tall, no swiping into form controls.
If inaccurate: replay to hear direction and magnitude	✓	Plays the gap, then the crossing time, back to back.
If accurate: optional follow-up on <i>how much</i> longer/shorter	✓	Opt-in checkbox. Student taps out the difference and hears their estimate against the true difference.
Track progress / show need for more instruction	✓	Every attempt logged; Progress screen and instructor report.

Actual Comparison Task (at the street)

Item	Status	Notes
First activation when a vehicle is detected	✓	
Direction: left = half street, right = full street	✓	Set before starting, so only one big control is needed while timing.
Second activation when the vehicle passes	✓	
Cancel — it turned off / never came	✓	Discards the trial; nothing is logged.
Immediate longer/shorter/same judgement	✓	Spoken, shown, sounded, and felt, then both durations replayed.
Ambient noise sampling to set a “quiet” threshold	✓	“Sample the quiet” records a 3-second median baseline. Measurements started more than 6 dB above it are flagged in the log and the report. Audio is analysed on-device and never recorded.
Flag samples taken in non-ideal conditions	✓	Flagged in the tally and in the instructor report.
Prompt to send data to instructor	✓	Progress → Share with my instructor.

Margin of error modes

Item	Status	Notes
Fixed mode (current)	✓	Default 0.40 s, editable.
Adaptive (behavioural shaping / ZPD)	✓	See the algorithm below.
Ceiling = the set margin	✓	
Hard-coded floor (50–100 ms suggested)	✓	0.10 s. Below that we would be measuring touchscreen latency, not time perception.
Rolling window of 10–15 trials of the same type	✓	10, per activity and street type.
≥80% → reduce margin — <i>“looking to you for help on the mathematics”</i>	✓	× 0.85 per step. Geometric, so each tightening is proportionally the same challenge. From 0.40 s it takes about 9 successful windows to reach the floor — a training arc, not one sitting.
<60% → increase margin	✓	Steps back to the last margin where they held ≥80%, capped at the ceiling.
60–80%	✓	Holds. Productive struggle.
Two successive failures → pin learner floor	✓	Floor pinned at the last ≥80% margin; instructor can clear it in Progress.
Record progress for the instructor	✓	Current margin per lane shown in Progress and in the report.

Data / Documentation

Item	Status	Notes
Submission of data to OMS (and user)		The report is generated and copied to the clipboard for the student to paste into any email or message. Direct send would need a backend or a mail integration — deliberately avoided for now.
Store OMS email, possibly user email		Student name is stored; instructor email is not yet.
Easily accessed method to send data		One button on the Progress screen.
Fully opt-in, preview before sending		The full report is shown verbatim first. Nothing leaves the device otherwise.
Include frequency, accuracy, progression, live-task data		All present, including noise flags.
Link from OMS that triggers a send prompt		Not built.
Self-voiced confirmation of what is being sent		The preview is screen-reader accessible, but there is no separate self-voicing.

Full accessibility

Item	Status	Notes
VoiceOver and TalkBack		Built for it throughout — landmarks, live regions, focus moved on screen change, labelled groups, large targets, no rotor needed. A real device pass is still outstanding (uc-j9n).
Keep flows easy for emerging skills; no rotor required		Every answer is a full-width button; no text fields in any practice flow.
Screen readers silent during replays		Announcements suppressed during every playback, in all four activities.
Reflow with zoom / large fonts		To 300%; every new screen collapses to a single column.
High contrast, respects user colour settings		Four themes plus high-contrast; gradients flattened where needed.
Battery considerations		No timers or animation loops run while idle, and the microphone is opened only on the “At the street” screen and released on leaving it. There is no explicit battery-saver toggle.

Still open

Ranked by how much they matter:

1. **Roadside audio check** — the sounds were rebuilt to fix a real clipping and speaker-response problem, but only verified in a browser and simulator. This needs someone standing next to traffic.
2. **VoiceOver/TalkBack device pass** — including the open question of whether double-tap-to-activate latency distorts a timing measurement.
3. **Instructor email stored + send link** — the last piece of the data-sharing flow.
4. **Per-street profiles** — save several streets by name rather than one set of times.
5. **Self-voicing** for consent confirmation.
6. **Android build** — the code path exists; the native project does not.

Open questions for Cindi

- **Adaptive step size:** is $\times 0.85$ per window the right aggressiveness? It is a single constant, trivially changed.
- **Learner floor:** should it ever expire on its own, or only be cleared by an instructor?
- **Gap sound:** continuous or two taps — which do clients find more intuitive? Both ship so this can be answered empirically.
- **Live task direction:** is left/right the right mapping to half/full street everywhere, or does it depend on the country and side of the road?
- **Measurement:** is averaging three walks right, or should outliers be discarded?